atomic structure

Name:

Class:

Date:

Time:

Marks: **71 marks**

Comments:

1. This question is about atomic structure.

(a) Atoms contain subatomic particles.

The table below shows properties of two subatomic particles.

Complete the table.

Name of particle	**Relative mass**	**Relative charge**
neutron		
		+1

(2)

An element **X** has two isotopes.

The isotopes have different mass numbers.

(b) Define mass number.

____

____

(1)

(c) Why is the mass number different in the two isotopes?

____

____

(1)

(d) The model of the atom changed as new evidence was discovered.

The plum pudding model suggested that the atom was a ball of positive charge with electrons embedded in it.

Evidence from the alpha particle scattering experiment led to a change in the model of the atom from the plum pudding model.

Explain how.

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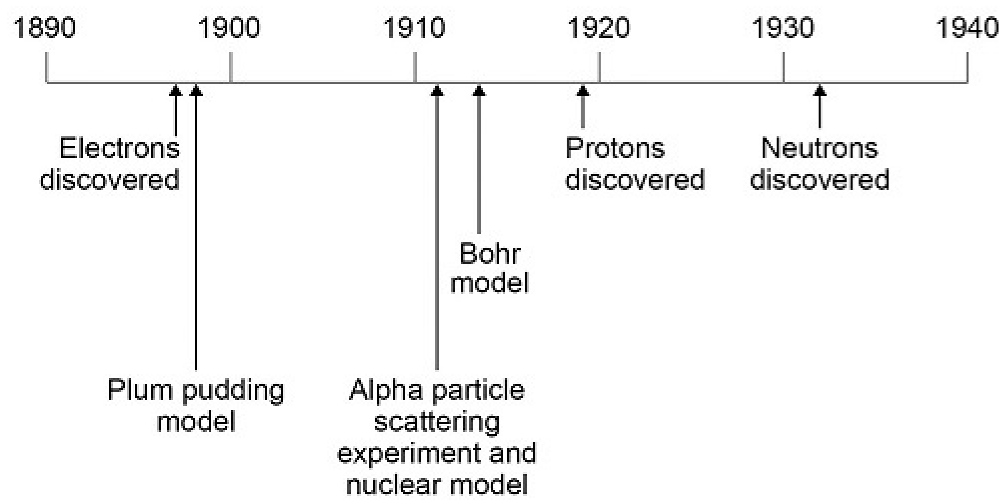
(4)
(Total 8 marks)

2. This question is about the development of scientific theories.

The diagram below shows a timeline of some important steps in the development of the model of the atom.



(a) The plum pudding model did not have a nucleus.

Describe **three** other differences between the nuclear model of the atom and the plum pudding model.

1 ______________________________

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2 ______________________________

____

____

3 ______________________________

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(3)

(b) Niels Bohr adapted the nuclear model.

Describe the change that Bohr made to the nuclear model.

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(2)

(c) Mendeleev published his periodic table in 1869.

Mendeleev arranged the elements in order of atomic weight.

Mendeleev then reversed the order of some pairs of elements.

A student suggested Mendeleev's reason for reversing the order was to arrange the elements in order of atomic number.

Explain why the student's suggestion **cannot** be correct.

Use the diagram above.

____

____

____

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(2)

(d) Give the correct reason why Mendeleev reversed the order of some pairs of elements.

____

____

(1)

(Total 8 marks)

3. This question is about atoms.

(a) What does the number 19 represent in ${}^{19}_{9}F$?

____

(1)

(b) How many atoms are present in one mole of fluorine atoms?

Tick (✓) **one** box.

2.03×10^{26} ☐

2.06×10^{23} ☐

6.02×10^{23} ☐

6.02×10^{26} ☐

(1)

(c) The plum pudding model of the atom was replaced by the nuclear model.

The nuclear model was developed after the alpha particle scattering experiment.

Compare the plum pudding model with the nuclear model of the atom.

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(4)

(d) An element has three isotopes.

The table shows the mass numbers and percentage of each isotope.

	Isotope 1	Isotope 2	Isotope 3
Mass number	24	25	26
Percentage (%)	78.6	10.1	11.3

Calculate the relative atomic mass (A_r) of the element.

Give your answer to 3 significant figures.

___

___

___

___

Relative atomic mass = ____________________

(2)

(Total 8 marks)

4. (a) Dmitri Mendeleev was one of the first chemists to classify the elements by arranging them in order of their atomic weights. His periodic table was published in 1869.

How did Mendeleev know that there must be undiscovered elements **and** how did he take this into account when he designed his periodic table?

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___

(2)

(b) By the early 20th century protons and electrons had been discovered.

Describe how knowledge of the numbers of protons and electrons in atoms allow chemists to place elements in their correct order and correct group.

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(3)

(c) The transition elements are a block of elements between Groups 2 and 3 of the periodic table.

(i) Transition elements have similar properties.

Explain why, in terms of electronic structure.

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____

____

____

(2)

(ii) There are **no** transition elements between the Group 2 element magnesium and the Group 3 element aluminium.

Give a reason why, in terms of electronic structure.

____

____

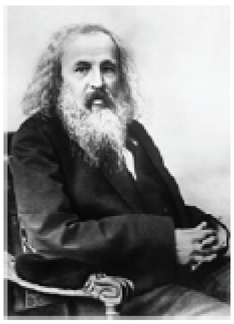
____

(1)

(Total 8 marks)

5. By 1869, about 60 elements had been discovered.
Mendeleev arranged these elements in a table, in order of their atomic weight.
He also put elements with similar chemical properties in the same columns.

Mendeleev and part of his table are shown below.



	Group							
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Period 1	H							
Period 2	Li	Be	B	C	N	O	F	
Period 3	Na	Mg	Al	Si	P	S	Cl	
Period 4	K Cu	Ca Zn	– –	Ti –	V As	Cr Se	Mn Br	Fe Co Ni

(a) (i) Name **one** element in Group 1 of Mendeleev's table that is not in Group 1 of the periodic table on the Data Sheet.
Give a reason why this element should not be in Group 1.

Name of element ___________________________

Reason ____

____

(2)

(ii) Which group of the periodic table on the Data Sheet is missing from Mendeleev's table?

____

(1)

(b) The gaps (–) in Mendeleev's table were for elements that had not been discovered.

(i) Compare Mendeleev's table with the periodic table on the Data Sheet.

Name **one** of the elements in Period 4 that had not been discovered by 1869.

____

(1)

(ii) Mendeleev was able to make predictions about the undiscovered elements. This eventually led most scientists to accept his table.

Suggest what predictions Mendeleev was able to make about these undiscovered elements.

(2)

(c) In terms of their electronic structure:

(i) state why lithium and sodium are both in Group 1

(1)

(ii) explain why sodium is more reactive than lithium.

(3)

(Total 10 marks)

6. An atom of aluminium has the symbol $^{27}_{13}Al$

(a) Give the number of protons, neutrons and electrons in this atom of aluminium.

Number of protons ______________________

Number of neutrons ______________________

Number of electrons ______________________

(3)

(b) Why is aluminium positioned in Group 3 of the periodic table?

____

____

(1)

(c) In the periodic table, the transition elements and Group 1 elements are metals.

Some of the properties of two transition elements and two Group 1 elements are shown in the table below.

	Transition elements		**Group 1 elements**	
	Chromium	Iron	Sodium	Caesium
Melting point in °C	1857	1535	98	29
Formula of oxides	CrO Cr_2O_3 CrO_2 CrO_3	FeO Fe_2O_3 Fe_3O_4	Na_2O	Cs_2O

Use your own knowledge **and** the data in the table above to compare the chemical and physical properties of transition elements and Group 1 elements.

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____

____

(6)

(Total 10 marks)

7. Use the periodic table and the information in the table below to help you to answer the questions.

The table shows part of an early version of the periodic table.

Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7
H						
Li	Be	B	C	N	O	F
Na	Mg	Al	Si	P	S	Cl

(a) Hydrogen was placed at the top of Group 1 in the early version of the periodic table.

The modern periodic table does **not** show hydrogen in Group 1.

(i) State one **similarity** between hydrogen and the elements in Group 1.

____

____

(1)

(ii) State one **difference** between hydrogen and the elements in Group 1.

____

____

(1)

(b) Fluorine, chlorine, bromine and iodine are in Group 7, the halogens.

The reactivity of the halogens decreases down the group.

Bromine reacts with a solution of potassium iodide to produce iodine.

$$Br_2 + 2KI \longrightarrow 2KBr + I_2$$

(i) In the reaction between bromine and potassium iodide, there is a reduction of bromine to bromide ions.

In terms of electrons, what is meant by reduction?

____

____

(1)

(ii) Complete the half equation for the oxidation of iodide ions to iodine molecules.

$$2I^{-} \qquad \longrightarrow$$

(2)

(iii) Explain, in terms of electronic structure, why fluorine is the most reactive element in Group 7.

____

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____

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____

____

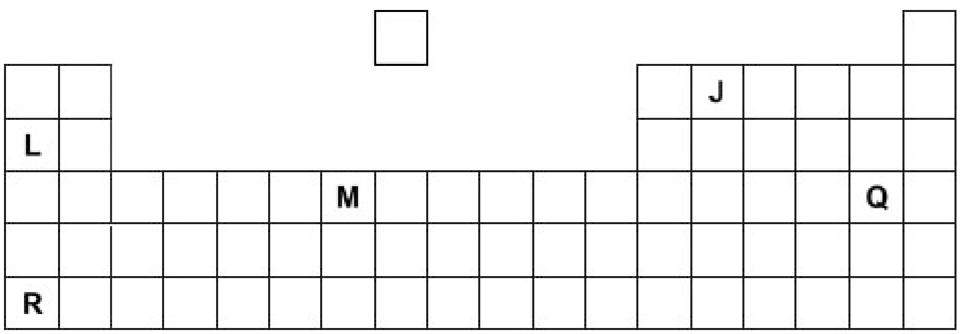
____

(3)

(Total 8 marks)

8. **Figure 1** shows an outline of the modern periodic table.

Figure 1



J, **L**, **M**, **Q** and **R** represent elements in the periodic table.

(a) Which element has four electrons in its outer shell?

Tick (✓) **one** box.

J ☐ **L** ☐ **M** ☐ **Q** ☐ **R** ☐

(1)

(b) Which **two** elements in **Figure 1** are in the same period?

__________________ and __________________

(1)

(c) Which element reacts with potassium to form an ionic compound?

Tick (✓) **one** box.

J ☐ **L** ☐ **M** ☐ **Q** ☐ **R** ☐

(1)

(d) Which element forms ions with different charges?

Tick (✓) **one** box.

J ☐ **L** ☐ **M** ☐ **Q** ☐ **R** ☐

(1)

(e) Which element has three electron shells?

Tick (✓) **one** box.

J ☐ **L** ☐ **M** ☐ **Q** ☐ **R** ☐

(1)

(f) In the 1860s scientists were trying to organise elements.

Figure 2 shows the table published by John Newlands in 1865.

The elements are arranged in order of their atomic weights.

Figure 2

H	Li	Be	B	C	N	O
F	Na	Mg	Al	Si	P	S
Cl	K	Ca	Cr	Ti	Mn	Fe
Co,Ni	Cu	Zn	Y	In	As	Se
Br	Rb	Sr	Ce,La	Zr	Di,Mo	Ro,Ru
Pd	Ag	Cd	U	Sn	Sb	Te

Figure 3 shows the periodic table published by Dmitri Mendeleev in 1869.

Figure 3

H							
Li	Be	B	C	N	O	F	
Na	Mg	Al	Si	P	S	Cl	
K Cu	Ca Zn	? ?	Ti ?	V As	Cr Se	Mn Br	Fe Co Ni
Rb Ag	Sr Cd	Y In	Zr Sn	Nb Sb	Mo Te	? I	Ru Rh Pd

Mendeleev's table became accepted by other scientists whereas Newlands' table was not.

Evaluate Newlands' and Mendeleev's tables.

You should include:

- a comparison of the tables
- reasons why Mendeleev's table was more acceptable.

Use **Figure 2** and **Figure 3** and your own knowledge.

(6)
(Total 11 marks)

Mark schemes

1. (a) (neutron) 1 0

both needed

allow (neutron) 1 neutral

1

proton 1 (+1)

both needed

1

(b) number of protons plus neutrons

allow number of protons and neutrons

ignore protons and neutrons unqualified

*do **not** accept references to mass or relative mass of protons and / or neutrons*

1

(c) (the isotopes contain) different numbers of neutrons

1

(d) most (alpha) particles passed (straight) through (the gold foil)

1

(so) the mass of the atom is concentrated in the nucleus / centre
or
(so) most of the atom is empty space

1

some (alpha) particles were deflected / reflected

1

(so) the atom has a (positively) charged nucleus / centre

if not awarded for MP2 allow (so) the mass of the atom is concentrated in the nucleus / centre.

1

[8]

2. (a) any **three** from: (nuclear model)

- mostly empty space

  allow the plum pudding model has no empty space

  allow the plum pudding model is solid

- the positive charge is (all) in the nucleus

  allow in the plum pudding model the atom is a ball of positive charge (with embedded electrons)
  *do **not** accept reference to protons*

- the mass is concentrated in the nucleus

  allow in the plum pudding model the mass is spread out
  *do **not** accept reference to neutrons*

- the electrons and the nucleus are separate

  allow in the plum pudding model the electrons are embedded

  allow in the nuclear model the electrons are in orbits

3

(b) electrons orbit the nucleus

*do **not** accept reference to protons / neutrons*
allow electrons are in energy levels around the nucleus
or
allow electrons are in shells around the nucleus

1

electrons are at specific distances from the nucleus

1

(c) atomic number is the number of protons

1

(and) protons were not discovered until later

ignore electrons / neutrons were not discovered until later

1

(d) so their properties matched the rest of the group

allow converse

1

[8]

3. (a) mass number

allow the number of protons + neutrons

1

(b) 6.02×10^{23}

1

(c) **Level 2 (3-4 marks):**
Scientifically relevant features are identified; the ways in which they are similar / different is made clear.

Level 1 (1-2 marks):
Relevant features are identified and differences noted.

Level 0
No relevant content.

Indicative content

similarities
- both have positive charges
- both have (negative) electrons
- neither has neutrons

differences

plum pudding model	**nuclear model**
ball of positive charge (spread throughout)	positive charge concentrated at the centre
electrons spread throughout (embedded in the ball of positive charge)	electrons outside the nucleus
no empty space in the atom	most of the atom is empty space
mass spread throughout	mass concentrated at the centre

4

(d) $$\frac{(24 \times 78.6) + (25 \times 0.1) + (26 \times 11.3)}{100}$$

or
(24 × 0.786) + (25 × 0.101) +
(26 × 0.113)

1

= 24.3

1

*an answer of 24.3 scores **2** marks*

[8]

4. (a) if placed consecutively, then elements would be in wrong group / have wrong properties
allow some elements didn't fit pattern

1

left gaps

1

(b) (elements placed in) atomic / proton number order

1

(elements in) same group have same number of <u>outer</u> electrons

1

any **one** from:

- number of protons = number of electrons
- reactions/(chemical) properties depend on the (outer) electrons
- number of shells gives the period

  allow number of shells increases down the group

1

(c) (i) (transition elements usually) have same / similar number of outer / 4th shell electrons

allow 2 electrons in outer shell

1

(because) inner (3rd) shell / energy level is being filled

ignore shells overlap

1

(ii) <u>2nd shell</u> / energy level can (only) have maximum of 8 electrons

accept no d-orbitals

or

<u>2nd shell</u> / energy level cannot have 18 electrons

1

[8]

5. (a) (i) *incorrect or no element = **0** marks*

hydrogen

allow H / H_2

1

all the other elements are metals

allow hydrogen is a not an (alkali / group 1) metal

ignore hydrogen is a gas

OR

copper (1)

allow Cu

(copper) is not an alkali metal (1)

allow Cu is a transition element / metal

allow any valid specific chemical property eg Cu does not react with water

ignore references to electronic structure

ignore physical properties

1

(ii) Group 0 / noble gases

ignore Group 8

1

(b) (i) scandium / gallium / germanium

accept Sc / Ga / Ge

allow Krypton / Kr

1

(ii) predicted they were metals

allow atomic mass / weight

ignore atomic structure

1

predicted their (chemical/physical) properties / reactivity

accept any chemical / physical property

allow similar properties if mentioned in context of a group

1

(c) (i) (both) have <u>one</u> / <u>an</u> electron in the outer energy level / shell

ignore form single plus ions

1

(ii) *accept shell for energy level*

accept converse explanation for lithium

if 'outer' not mentioned, max 2 marks

ignore sodium reacts more easily

sodium loses one outer electron <u>more</u> easily (than lithium)

1

because outer electrons/energy level furth<u>er</u> from the nucleus in sodium **or** because sodium has <u>more</u> shells (than lithium)

*do **not** accept 'more outer shells'*

allow sodium (atom) is larger

1

because forces/attraction to hold outer electron are weak<u>er</u> in sodium (than lithium)

accept more shielding in sodium (than lithium)

1

[10]

6. (a) 13 (protons)

The answers must be in the correct order.

*if no other marks awarded, award **1** mark if number of protons and electrons are equal*

1

14 (neutrons)

1

13 (electrons)

1

(b) has three electrons in outer energy level / shell

allow electronic structure is 2.8.3

1

(c) **Level 3 (5–6 marks):**
A detailed and coherent comparison is given, which demonstrates a broad knowledge and understanding of the key scientific ideas. The response makes logical links between the points raised and uses sufficient examples to support these links.

Level 2 (3–4 marks):
A description is given which demonstrates a reasonable knowledge and understanding of the key scientific ideas. Comparisons are made but may not be fully articulated and / or precise.

Level 1 (1–2 marks):
Simple statements are made which demonstrate a basic knowledge of some of the relevant ideas. The response may fail to make comparisons between the points raised.

0 marks:
No relevant content.

Indicative content

Physical
Transition elements

- high melting points
- high densities
- strong
- hard

Group 1

- low melting points
- low densities
- soft

Chemical
Transition elements

- low reactivity / react slowly (with water or oxygen)
- used as catalysts
- ions with different charges
- coloured compounds

Group 1

- very reactive / react (quickly) with water / non-metals
- not used as catalysts
- white / colourless compounds
- only forms a +1 ion

6

[10]

7. (a) (i) any **one** from:

- one electron in the outer shell / energy level
- form ions with a 1+ charge

1

(ii) any **one** from:

- hydrogen is a non-metal
- (at RTP) hydrogen is a gas
- hydrogen does not react with water
- hydrogen has only one electron shell / energy level
- hydrogen can gain an electron **or** hydrogen can form a negative / hydride / H^- ion
- hydrogen forms covalent bonds **or** shares electrons

accept answers in terms of the Group 1 elements

1

(b) (i) (bromine) gains electrons

it = bromine

*do **not** accept bromide ion gains electrons*

ignore loss of oxygen

1

(ii) I_2

must both be on the right hand side of the equation

1

$+ 2e^-$

*$2I^- - 2e^- \rightarrow I_2$ for **2** marks*

1

(iii) fluorine is the smallest atom in Group 7 **or** has the fewest energy levels in Group 7 **or** has the smallest distance between outer shell and nucleus

*the outer shell **must** be mentioned to score 3 marks*

1

fluorine has the least shielding **or** the greatest attraction between the nucleus and the outer shell

1

therefore fluorine can gain an electron (into the outer shell) more easily

1

[8]

8. (a) **J**

1

(b) **M** and **Q**

either order

1

(c) **Q**

1

(d) **M**

1

(e) **L**

1

(f) **Level 3 (5-6 marks):**
A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.

Level 2 (3-4 marks):
Some logically linked reasons are given. There may also be a simple judgement.

Level 1 (1-2 marks):
Relevant points are made. They are not logically linked.

Level 0
No relevant content

Indicative content

comparative points
- both tables have more than one element in a box
- both have similar elements in the same column
- both are missing the noble gases
- both arranged elements in order of atomic weight

advantages of Mendeleev / disadvantages of Newlands
- Newlands did not leave gaps for undiscovered elements
- Newlands had many more dissimilar elements in a column
- Mendeleev left gaps for undiscovered elements
- Mendeleev changed the order of some elements (e.g. Te and I)

points which led to the acceptance of Mendeleev's table
- Mendeleev predicted properties of missing elements
- elements with properties predicted by Mendeleev were discovered
- Mendeleev's predictions turned out to be correct
- elements were discovered which fitted the gaps

6

[11]

Examiner reports

1.

(a) Approximately 80% of students scored both marks for the names / properties of the proton and the neutron.

(b) Approximately 70% of students gave the correct definition of mass number.

However, there were some incorrect references to mass, for example, ‘mass number is the total mass of the protons and neutrons’.

(c) Approximately 80% of students correctly answered that isotopes of the same element have different numbers of neutrons.

(d) Most students made a good attempt at this question about the alpha particle scattering experiment.

The marks for the ideas that some particles were deflected and/or reflected and that there is a charged nucleus were the most frequently awarded.

The mark for the idea that <u>most</u> of the particles passed straight through the gold foil was the least frequently awarded.

Those students gained fewer marks tended to focus on expected outcomes but didn’t know exactly what happened in the experiment.

2.

(a) Students had difficulty in expressing their ideas clearly; electrons in the plum pudding model were variously described as ‘floating’ or ‘covering’ the atom rather than embedded in the atom. A popular misconception was that the plum pudding model had an overall positive charge. A large number of students lost credit by referring to protons and neutrons in the nuclear model, despite the timeline which showed they had not been discovered at the time of the development of the model. Some answers did not make it clear which model was being written about.

(b) Many students stated that the Bohr model had electrons that orbit the nucleus, but fewer said that the orbits were at specific distances from the nucleus. Again, it was not uncommon to see mention of protons and neutrons despite the timeline provided.

(c) Many students were able to deduce that the issue was that protons had not yet been discovered, but few also stated the relevance of the fact that the atomic number is the number of protons.

(d) Few students could express coherently that the reason was to place elements with similar properties in the same group. There was a lot of imprecise language which did not earn credit; ‘reactivity’ is not the same as properties and placing elements ‘with’ those with similar properties does not necessarily mean in the same group or column.

4.

(a) The great majority of students knew that Mendeleev left gaps in his table. Just over half left their answer at that, scoring one mark, but at least another third went on to indicate that the gaps were needed to maintain or restore a pattern in the periodic table.

(b) This question provided a fairly even spread of scores. Many answers were quite confused but nevertheless managed to gain some marks, the most frequent correct answer being the connection between outer electrons and group number.

(c) (i) This question proved difficult to most students. About one quarter gained the mark for the 4s electrons but only about a tenth of students managed to describe the filling of the 3d sub-shell adequately.

(ii) The great majority of students were baffled by this question, with only a few gaining a mark. Answers were often based on atomic number, in effect pointing out that there were no missing atomic numbers in Period 3, or stating that insertion of elements into a non-existent gap would require protons or electrons to be split.

5. (a) (i) This was well done by many of the students, although there were frequent references to electronic structure in the explanation. Copper was identified much more frequently than hydrogen and the most popular correct explanation was that it is a transition metal.

(ii) Approximately half of students answered correctly. The overwhelming majority of those who did not suggested the transition elements. There were a number who answered group 8 but frequently this was stated together with group 0 (i.e. Group 8/0) so the students were not penalised. A small number suggested the halogens.

(b) (i) Most students correctly identified one of the required elements. The most common wrong answers were from Group 4 (lead and tin) rather than Period 4, although there were also many seemingly random answers.

(ii) Both marking points (atomic weight and properties) were awarded frequently but most students failed to gain both marks, concentrating on one or the other. Anachronistic predictions concerning atomic structure such as atomic number or number of outer electrons were common. A significant number of students did not answer the question correctly, explaining instead how Mendeleev knew there were missing elements and why he left gaps. Presumably they did not read the question carefully enough.

(c) (i) Majority of students were awarded a mark. Commonly rejected answers included suggestions that both lithium and sodium had just one energy level (shell), that they had the same, but unspecified, number of outer electrons, or made reference to their ions or reactivity.

(ii) This was generally answered well. Most students are aware of the reasons for the difference in reactivity of lithium and sodium. Fewer students than normal omitted the comparative in one, two or all three of the statements. A few were penalised for not stating the word ‘outer’ in their answer. Some errors concerned the description of the electrostatic attraction as (electro)magnetic, gravitational, intermolecular, or as a bond.

7. (a) (i) The majority of students gained the mark for describing that hydrogen, like the alkali metals, has one electron in the outer energy level or shell.

(ii) The majority of students gained the mark, usually for stating that hydrogen is a gas or non-metal or has only one energy level or shell. Some students described hydrogen as unique in having no neutrons, without considering the difference between hydrogen and Group 1 metals.

(b) (i) This question was well answered, with most students appreciating the role of electrons in reduction.

(ii) Balancing the half equation proved demanding, with less than half of answers gaining full marks. The most common mistake was to show iodine as 2I, instead of I_2. Students were clearly confused regarding the effect of losing or gaining a negative electron, and often the charges did not balance.

(iii) Two-thirds of students scored two or three marks in this question. The most common response related to the size of the fluorine atom and the subsequent level of attraction between the nucleus and outer shell electrons. Many students were familiar with the concept of shielding, as it often appeared in the responses. The last mark was the one that was the least often gained. Some students did not realise that the attraction of an additional electron was important in the explanation.